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 Review the assignment due date

a11g-final-submission

Team Number: 32

Team Name: LumeEdge

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GitHub Repository URL: <https://github.com/ese5160/a11g-final-submission-s26-s26-t32-lumeedge>

GitHub WebPage URL: <https://ese5160.github.io/a11g-final-submission-s26-s26-t32-lumeedge>

1. Video Presentation

<https://youtu.be/i7tSZm3Seq0>

and on github webpage:

<https://ese5160.github.io/a11g-final-submission-s26-s26-t32-lumeedge>

2. Project Summary

Please see github webpage:

<https://ese5160.github.io/a11g-final-submission-s26-s26-t32-lumeedge>

3. Hardware & Software Requirements

3.1. Hardware Requirements Specification (HRS)

The following requirements define the electrical and physical constraints of the LumeEdge system. They are prioritized to ensure the device meets the **ESE 516** graduate-level complexity standards while remaining strictly testable.

3.1.1 Power System Requirements

ID	Requirement Description	Priority	Validation
HRS-01	The device shall be powered by a single-cell Lithium-Ion battery with a nominal voltage of 3.7V.	Mandatory	Passed. It can run in battery mode
HRS-02	The power system shall include an charge management controller to facilitate battery charging via USB-C.	Mandatory	Passed. It can charge the battery
HRS-03	The system shall generate a regulated 3.3V rail ($\pm 5\%$) using a Buck Boost Converter topology to	Mandatory	Passed. We measure it with multimeter

ID	Requirement Description	Priority	Validation
	power the MCU and sensors efficiently.		
HRS-04	The device should auto switch to external	<i>Recommended</i>	Passed. It will use USB-C if connected

3.1.2 Computing & Communication Requirements

ID	Requirement Description	Priority	Validation
HRS-05	The primary microcontroller shall be the SIWG917Y121MGABA SoC.	Mandatory	Passed.
HRS-06	The microcontroller shall utilize Wi-Fi to maintain connectivity with the Cloud LLM gateway.	Mandatory	Passed. Wifi works normally and can communicate with gateway.
HRS-07	The system shall utilize a dedicated I^2C bus operating at a minimum of 100 kHz (Standard Mode) for sensor communication.	Mandatory	Passed. I2C is in standard mode.

3.1.3 Sensing & Actuation Requirements

ID	Requirement Description	Priority	Validation
HRS-08	The system shall include a Ambient Light Sensor to measure illuminance values.	Mandatory	Passed. It has one.
HRS-09	The system shall include a Environmental Sensor to measure Temperature ($\pm 1.0^\circ C$) and Relative Humidity ($\pm 3\%$).	Mandatory	Passed. It has one. Accuracy is $\pm 0.2C$ and $\pm 2\%RH$ in datasheet.

ID	Requirement Description	Priority	Validation
HRS-10	The device shall control the LED actuator brightness using Pulse Width Modulation (PWM) with a frequency of no less than 480 Hz to prevent visible flicker.	Mandatory	Passed. Frequency is 5KHz.
HRS-11	The system should have a screen to display system status.	Recommended	Passed. It has one.
HRS-12	The actuator shall have a driver to control a motor.	Mandatory	Passed. It has one.

3.1.4 Physical & Budgetary Requirements

ID	Requirement Description	Priority	Validation
HRS-13	The total Bill of Materials (BOM) cost for the sensing and actuation subsystem should not exceed \$30.00 USD.	Recommended	Passed.
HRS-14	The PCB should be designed with mounting holes compatible with a standard electrical wall box enclosure.	Recommended	Passed.
HRS-15	All major components (MCU, Regulators, Sensors) should be Surface Mount Devices (SMD) to reflect professional manufacturing standards.	Recommended	Passed.

3.2. Software Requirements Specification (SRS)

3.2.1 System Initialization & Configuration

ID	Requirement Description	Priority	Validation
SRS-01	The system shall provide a mechanism (e.g., SoftAP or BLE provisioning) to accept and save	Mandatory	Not Passed. We use locally auto

ID	Requirement Description	Priority	Validation
	local Wi-Fi SSID and Password credentials.		switching mechanisms instead.

3.2.2. Sensor Data Acquisition

ID	Requirement Description	Priority	Validation
SRS-02	The firmware shall poll the Light Sensor every 1 seconds (configurable) to obtain the current Lux value.	Mandatory	Passed. About every 75ms.
SRS-03	The firmware shall poll the SHT40 sensor every 5 seconds to obtain Temperature and Humidity data.	Mandatory	Passed. About every 550ms.

3.2.3 Cloud & LLM Integration

ID	Requirement Description	Priority	Validation
SRS-04	The device shall establish a secure (TLS/SSL) connection to the Cloud Gateway (e.g., AWS IoT or Azure IoT).	Mandatory	Passed. We use azure.
SRS-05	The system shall format sensor telemetry into a JSON payload containing <code>{"lux": value, "temp": value, "user_intent": string}</code> for transmission.	Mandatory	Passed.
SRS-06	The system shall parse incoming JSON commands from the Cloud LLM containing target brightness levels (0-100%) and motor speed (if supported).	Mandatory	Passed.
SRS-07	If the Wi-Fi connection is lost, the device shall maintain the last known actuator state until connectivity is restored.	Mandatory	Half Passed. the device maintain the last known actuator

ID	Requirement Description	Priority	Validation
			state, but reboot is needed to reconnect.

3.2.4 Actuation & Control Logic

ID	Requirement Description	Priority	Validation
SRS-08	The firmware shall generate a hardware PWM signal on the GPIO connected to the MOSFET gate.	Mandatory	Passed.
SRS-09	The control loop shall implement a "Soft-Ramp" feature, changing brightness by no more than 10% per 100ms, to prevent jarring transitions.	Mandatory	Passed. max rate is 1%/1s
SRS-10	The system should implement a PID control algorithm to maintain target brightness if external lighting conditions change rapidly.	<i>Recommended</i>	Passed. P = 0.001, I=D=0

4. Project Photos & Screenshots

Please see github webpage:

<https://ese5160.github.io/a11g-final-submission-s26-s26-t32-lumeedge>

5. Codebase

Do *not* commit any of your source code to this repository. Rather, provide links to the other GitHub repository you've already been using with your firmware.

- A link to your final embedded C firmware codebases:
https://github.com/ese5160/final-project-firmware-s26-t32-lumeedge/tree/main/final/wifi_http_otaf_soc

- A link to your Node-RED dashboard code: <https://github.com/ese5160/final-project-firmware-s26-t32-lumeedge/tree/main/Node-RED>
- Links to any other software required for the functionality of your device